Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/06/2005
PI Name:	Shapiro, Irving M. Ph.D.		
Project Title:	Effect of microgravity on human osteoblast life history		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Irving.Shapiro@jefferson.edu	**Fax:**	FY 215-955-9159
PI Organization Type:	UNIVERSITY	**Phone:**	215-955-7217
Organization Name:	Thomas Jefferson University		
PI Address 1:	1015 Walnut St		
PI Address 2:			
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19107	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	06/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1400		
Performance Goal No.:			
Performance Goal Text:			

Adaptation of bone cells to microgravity remains poor despite long term spaceflight experience. Histomorphometric studies of osteopenic bone point to microgravity affecting bone loss through a modest increase in resorption and a decrease in bone formation. In the context of this proposal, we propose to examine the impact of microgravity on terminal activities of osteoblasts, namely the induction of the apoptotic response. The major hypothesis to be tested is that microgravity promotes normal osteoblast development and induces bone cell apoptosis. There are three Specific Aims. The objective of Specific Aim 1 is to quantify the effects of microgravity on human bone cells. For this purpose the cells will be seeded onto hollow biodegradable microcarriers and maintained in rotating bioreactor that was designed by NASA for tissue engineering. Microgravity conditions will be imposed on the cells using interactive numerical modeling techniques that have been developed by our group. Development of the osteoblast culture will be assessed by measuring the activity of alkaline phosphatase, the formation of apatite and the expression of genes that characterize the

Task Description:

bone cell phenotype. The second Specific Aim is directed at determining whether microgravity influences osteoblast function in terms of modulating bone cell proliferation and osteoblast death. Using a variety of techniques we will determine whether the induction of death is through activation of the apoptotic response. To understand the death pathway, we will then examine the impact of microgravity on expression of members of the Bcl-2 family of genes. We will test the hypothesis that microgravity causes activation of those genes that lead to apoptosis and a decrease in the expression of genes that protect cells from the apoptotic response. The last Specific Aim is to measure Ca2+ and Pi accumulation at the microcarrier under microgravity conditions and to determine whether the accumulated ions enhance osteoblast apoptosis. We will investigate whether microgravity causes an accumulation of ions that promote human bone cell apoptosis.Numerical modeling will compliment the experimental measurements by providing the surface rate accumulation of ions under conditions of simulated microgravity. Specific Aim 1. To quantify the effects of microgravity on human bone cells in the new microcarrier system in terms of (a) expression of the genes that are characteristic of bone (Osf-2, osteocalcin, osteopontin and type I collagen); (b) the presence of biological apatite (c) the activity of alkaline phosphatase
Hypothesis 1: microgravity modulates the expression of major osteogenic genes, the synthesis of the mineralized phase of bone and the maturation of the bone cell phenotype.

Numerical modeling: will provide information on hydrodynamic variables that govern shear stress at the cell surface under microgravity conditions; it will be used to generate data on the optimum location of cells in the rotating wall vessel with adequate information on nutrient levels and pO2 .

Specific Aim 2. To examine evaluate the effects of microgravity on osteoblast function in terms of bone cell proliferation and cell death. To ascertain whether microgravity induces human bone cell apoptosis.

Hypothesis 2: microgravity induces bone cell death by enhancing the apoptotic process.

Numerical modeling: same as Aim 1.

Specific Aim 3. To measure Ca2+ and Pi accumulation at the microcarrier under microgravity conditions and to determine whether the accumulated ions enhance osteoblast apoptosis.

Hypothesis 3: To investigate whether microgravity causes an accumulation of ions that promote human bone cell apoptosis.

Numerical modeling: numerical modeling will compliment the experimental measurements by providing the surface rate accumulation of ions under conditions of simulated microgravity. The other aspects of numerical modeling remain the same as Aim 1. Knowledge of the adaptation of the human bone cells to microgravity remains poor despite long term space flight experience and the availibilty and accurate techniques for bone mass measurements. Microgravity has been noted to modify the function of bone cells and disturb metabolism. Cellular changes are reported to oocur early and if prolonged result in osteopenia. Bone loss is thought to be due to a modest loss of bone through resorption and a decrease in bone formation. Some studies suggest that reveresibilty of bone loss is unlikely after six months of immobilization or weightlessness. For this reason, prolonged weightloss of astronauts may present problems on return to Earth with increased risk of fractures and premature osteoporosis. Thus, there is a great need to gain to a fundamental understanding of the effects of microgravity on human bone cell function. In the context of this investigation, we are examining the impact of microgravity on terminal activities of osteoblasts, namely the induction of the apoptotic response. Our major hypothesis is that microgravity promotes bone cell apoptosis.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

BIOCHEMISTRY AND CELL BIOLOGY STUDIES
Effects of Vector Averaged Gravity on Osteoblasts and Osteoclasts We investigated the hypothesis that microgravity directly induces osteoblast apoptosis, using clinorotation (vector-averaged gravity) to simulate aspects of microgravity. Cognizant of the drawbacks associated with the use of conventional clinostats, we designed a novel system in which osteoblasts were embedded in alginate carriers. By using both non-rotationally stabilized and rotationally stabilized carriers to culture cells in normal gravity conditions in the HARV, we were able to control for extraneous factors such as fluid flow-induced shear stress, vibration, and convective mixing. We fabricated two types of alginate beads to support osteoblast culture in the HARV: rotationally stabilized and non-rotationally stabilized carriers, designated polar and nonpolar carriers, respectively. Polar carriers were created with a buoyant pole that imparts rotational stability.Nonpolar carriers spun about their own axes in the HARV while polar carriers remained in a constant orientation. Accordingly, as the HARV rotates, cells in nonpolar carriers were subjected to vector-averaged gravity while cells in polar carriers remain in normal gravity conditions.

The influence of vector-averaged gravity on osteoblast survival was evaluated using three assays. First, following culture in vector-averaged gravity for 1, 5, or 14 days, the MTT assay was used to assess cell survival. The viability of cells cultured in vector-averaged gravity was the same as cells maintained in normal gravity, even extended cultures of 14 days with reduced serum. Second, following culture, the cells were stained with CellTracker (5-chloromethylfluorescein diacetate), an indicator of intracellular glutathione S-transferase activity of the cultured cells. No difference was observed in the activity of this enzyme and by inference the thiol redox status of the cells. Third, a challenge with two apoptogens failed to show any difference in apoptotic sensitivity between the cells maintained in normal or vector-averaged gravity conditions. None of these studies provided any support for the hypothesis that vector-averaged gravity affects osteoblast survival. In contrast to other studies, we found that vector-averaged gravity, even for extended time periods, did not reduce osteoblast viability (MC3T3-E1 cells). Furthermore, osteoblasts were not sensitized to apoptogens by exposure to vector-averaged gravity. These findings suggest that vector-averaged gravity alone is insufficient to induce or modulate the susceptibility of osteoblasts to apoptosis.

We also evaluated the effects of vector-averaged gravity on the differentiation status of the cells by assessing the expression of several genes characteristic of mature osteoblasts. RT-PCR analysis revealed an increase in expression of osteocalcin in vector-averaged gravity, although other markers of differentiation remained unchanged. This finding is in agreement with those of Hughes-Fulford et al. who showed similar effects on osteocalcin expression in the same cell line (MC3T3) flown in space. Although osteocalcin expression is normally associated with the expression of other genes

Task Progress:	involved in matrix mineralization, we did not detect measurable changes in the expression of other extracellular matrix proteins (OPN, COL1) or RUNX2. We increased the effective gravity by subjecting cells to hypergravity using a modified centrifugation technique. Osteoblasts in alginate carriers were cultured in a centrifuge (Centra CL3R, IEC). The centrifuge was coupled with an incubator to maintain 37C and 5% CO2 and custom programmed to operate at a minimum of 200 rpm. By culturing cells at different radial positions, two RCF values were evaluated in a single experiment (3 g and 6 g). While no changes in viability were observed due to vector averaged gravity, preliminary results using the centrifuge system suggest the hypergravity decreases the viability of osteoblasts. Use Of The New Modified HARV System To Study Bone Resorption A further modification of the carrier system is the incorporation of a bone into the alginate carrier. Intact adult mouse distal phalanges were subjected to simulated microgravity in the HARV for 5 days. The new system has the advantage that we can now evaluate the mineralization status of intact bones subjected to microgravity. To evaluate and compare the effect of microgravity and unit gravity on osteoclast formation, the expression of RANK and cathepsin K were determined by mRNA in situ hybridization. An increase in RANK expression was observed after 5 days of exposure to microgravity. The level of RANK expression was elevated above values observed in bones exposed to unit gravity. Similarly, the level of cathepsin K expression was elevated in microgravity as compared to unit gravity. In addition, the number of osteoclasts present in bones that had been exposed to microgravity was determined by enzyme histochemistry. There was an increase in the number of cells that expressed TRAP in bones exposed to microgravity as compared with bones in normal gravity. The intensity of the stained cells was also increased. NUMERICAL STUDIES ON PHOSPHATE KINETICS AND HARV DYNAMICS Numerical Modeling Of Phosphate Ion Transfer In The Harv: Application Of Michaelis- Menten Kinetics For Phosphate Ion Consumption By Napi Co-Transporter System We have modeled the transport and accumulation of the phosphate ions in a basic multicellular unit consisting of osteoclasts and osteoblasts situated adjacent to each other in straining flows. Two such flows are considered; One corresponding to shear levels representative of trabecular bone conditions at normal gravity, and the other corresponding to shear level that is representative of simulated microgravity conditions prevailing in a NASA designed Rotating Wall Bioreactor (RWV) culture. The consumption of Pi by osteoblasts is assumed to follow Michaelis-Menten (MM) kinetics aided by NaPi co-transporter system. MM kinetics views the NaPi co-transporter as a system for transporting extracellular Pi into the osteoblast. Our results show, for the condition investigated here, the net accumulation of the phosphate ions at the osteoblast surface at simulated microgravity conditions is higher by as much as a factor of 4. Such increased accumulation may lead to enhanced apoptosis and may help explain the increased bone loss observed in microgravity. Numerical Modeling Of Particulate Motion In The HARV: Relevance Of The Rotating Frame Of Reference Detailed numerical simulation results for the three-dimensional motion of a single or many spherical rigid particles in a Newtonian-fluid filled cylindrical vessel rotating about its horizontal axis are provided. The ranges of low and intermediate particle Reynolds number have been investigated. We have developed a new DNS based on the Arbitrary Lagrangian-Eulerian (ALE) moving grid unstructured finite element scheme and is described in a frame of reference with its origin coinciding with the center of the cylinder and rotating with the fluid (a rotating frame of reference). The newly developed model is noted to provide an accurate description of the flow and of the particulate behavior. Effecting the solution in the rotational frame of reference is seen to be computationally very fast and efficient for this problem compared to conventional computations in an inertial frame of reference. Several novel features in regard to two particle interactions in rotating flows have been revealed by this detailed numerical study. EMPIRICAL DATA ON ALGINATE CARRIER-HARV SYSTEM A novel apparatus, based on a previously reported method, was used to encapsulate osteoblast-like cells in alginate carriers. Carriers containing MC3T3 cells were cultured together in a 50 ml re-usable HARV (Synthecon). Prior to loading the HARV with carriers, we used Ficoll 400 (Sigma) to increase the density of the culture medium. The buoyancy of the carriers minimized collisions between the alginate particles and the walls of the HARV. We evaluated the trajectory of the carriers in the HARV by video and video microscopy using a camera (XL1S, Canon) with a macro lens. The camera tracked the motion and orientation of carriers in the HARV. The vector-averaged gravity experienced by the cells was calculated using the cosine of the angular position of the carriers. These results correlated very well with the predictions made by numerical simulations. Over the course of one revolution, the cells in polar and nonpolar carriers experience time-averaged gravitational accelerations of 0.993g (+/-0.005), and 0.088g (+/-0.084) respectively.
Bibliography Type:	Description: (Last Updated: 05/15/2006)